

Module 2 - Timeseries analysis

International Relations and Transboundary Cooperation

As an engineer you are called from the Swiss federal government to perform a preliminary evaluation on the impact of the Ill River regulation on the Rhein hydrological and suspended sediment regime.

The Ill River is the last major tributary of the Rhein before it enters Lake Constance. The Ill River catchment is entirely in Austria and its discharge at the confluence with the Rhein is partially regulated by the Illspitz Kraftwerk.

You are given discharge and suspended sediment concentration time series from the *Gisingen* ([eHYD](#)) measuring station on the Ill River and of the *Diepoldsau, Rietbrücke* ([BAFU](#)) station for the Rhein River, downstream of the confluence.



Data:

Gisingen

- 2003-2021: C suspended sediment concentration (g/L – 15min)
- 1976-2013: Q discharge (m³/s – 15min)

Diepoldsau, Rietbrücke

- 2012-2025: C suspended sediment concentration (g/L – 10 min)
- 1984-2025: Q discharge (m³/s – 10 min)

Analysis:

1. Timeseries review: [2]

Compute the monthly mean value for both Q and C time series, you will be working with monthly data aggregates. Plot the timeseries in their whole length, both discharge (Q) and concentration (C). Do they look stationary? Do you notice trends?

Test for stationarity. Build a linear regression model and test the significance of the slope to 5% significance. Subtract the mean or any statistically significant linear trend. The resulting timeseries should have a mean of zero and a finite variance σ_Z^2 .

Comment whether the statistical trend in the discharge data is the result of a measuring error or a physical underlying process. Express your opinion based on scientific evidence, if there is any.

2. Timeseries Modelling: [5]

Compute the empirical autocorrelation and the approximate 95% confidence interval. Compute a partial autocorrelation and the approximate 95% confidence interval.

Fit an AR and an ARMA model based on the results of the autocorrelation analysis. Justify your choices of the order based on quantitative observations. Can you use the same model order for Q at both locations? Answer the same question also for C. Either if yes or no, discuss the choice and the possible reasons why, provide a qualitative answer.

3. Timeseries application & evaluation: [5]

Compute and plot the empirical autocorrelation function (ACF) of the observed data and the theoretical ACF implied by each fitted model. Display them on the same plot together with the approximate 95% confidence intervals

Compute and plot the ACF of the residuals for each model, including the same 95% confidence intervals. Verify that all residual autocorrelations lie within the confidence bounds, discuss whether any significant temporal dependence remains.

Test the independence of the residuals at the 5% significance level with a Porte-Manteau Test. Test the normality of each residual series at the 5% significance level with a probability plot and PPCC test.

Discuss your analysis so far. What is the most appropriate model and why? (AR or ARMA?). Once you have chosen the most appropriate model (one for C and one for Q), discuss if the order of the model is chosen such that the model is parsimonious, i.e.: balanced between performance and complexity. Discuss your choice and, if reasonable, adjust the order of the model.

4. Ill to Rhein relative sediment influence: [5]

For the chosen model, generate 10 time series for the next 10 years. Plot the synthetic timeseries with the normalized data series (do not add back the mean or the trend for comparison with historical data). Compare the statistical characteristics.

DISCLAIMER: The objective is to compare long-term statistical properties rather than time-synchronous behaviour. That is why we could calibrate the models on the full available time series at each station. Discuss the implications of comparing non-synchronous time series when estimating sediment contributions

At each station, calculate the sediment mass $M(t) = C(t) \cdot Q(t)$ [kg/s]. Compute the average monthly and yearly sediment yields. Answer the following questions: Do the synthetic times series produce similar mass yields (monthly and yearly averages), at each station? How much does the Ill contribute to the monthly and yearly sediment mass yields at Diepoldsau? Do the synthetic timeseries provide valid estimates of the relative sediment contribution of the Ill to the Rhein River?

5. Independent variables? :[3]

Is it appropriate to simulate discharge and suspended sediment separately?

Estimate the correlation between the two variables and both locations. Plot the joint distribution of (Q, C) and describe the relationship between the two variables. Can Q and C be considered statistically independent? Produce quantitative evidence to support the answer to the previous question.

Discuss the shortfalls of the methodology of simulating dependent variables with independent models? How would you resolve these shortcomings?

Method:

The final product of this exercise will be automatic code as a Jupyter notebook in *.ipynb* format to perform the analysis. Each of the steps (1 to 5) must be in a different section of the Jupyter notebook. Each section must be organized in the following way:

- Markdown cell: Title
- Cell (MAIN) where you run the analysis
- Cell (PLOT) following the main cell, where you plot the results
- Cell (PRINT) to print the results of the analysis, in nice format
- Markdown cell to comment on the results

At the start of the notebook import all the necessary packages and libraries for the analysis.

For each section of the *.ipynb* file, write the necessary functions in a separate *.py* file, import the relevant functions from the *.py* at the start of each section. In the Jupyter Notebook keep ONLY the MAIN, PLOTS and Comments to the results. In the separate *.py* files, each function must be described as (1) what it does, (2) input (3) outputs. Inputs and outputs described as type of variable and meanings.

The code for Module 2 exercise must be running from start to finish. I suggest making each section independent from the others, not compromising the entire output. A non-running code will not be graded.

Module 2 is a total of 20 points, each section with the score assigned in brackets []

GitHub Copilot

The objective of this module is to get used to version control workflow. In professional engineering, teams do not work on the same file simultaneously. Instead, you will work separately on features, subject your work to peer review, and only integrate it into the stable production code once it has been approved. Groups are composed by two people, here referred to A and B. I suggest the following workflow:

1. Repository Setup: One group member (A) creates the GitHub repository. Initialize it with the principal Jupyter Notebook and the all the files .py for each of the project sections. Invites member (B)
2. Branching: Never push code directly to the main branch. For every task, such as Section 1, create a new Branch. For example, create the branch "Timeseries review:" and work in there until the section is complete.
3. Commits: Commit your changes often. Each commit should be a small, logical step, for example: "*implement Kendall-Theil regression*".
4. The Pull Request: When member A completes a section, it opens a Pull Request on GitHub. This is the signal that the code is ready to be inspected.
5. Peer Review: Member B reviews the code in the Pull Request. Member B leaves comments on specific lines of code suggesting improvements or pointing out bugs.
6. Refinement: Member A addresses the review comments, pushes the necessary fixes to their branch, and notifies Member B. Once the code is perfect, Member B can close the Pull Request and proceed into the merging.
7. The Merge: Only after approval is the branch merged into main. Once merged, delete the feature branch to keep the repository clean.

Best Practices

Logic vs. Display: Keep all math and functions in .py files; use. ipynb only for visualization. Jupyter Notebooks are notoriously difficult to merge in Git. .py files are clean text.

Main Protection: The main branch must always be functional. If the code on main does not run from start to finish, the project is considered broken.

>> Conflict Resolution: Merge main into your feature branch before your final merge. This forces you to solve merge conflicts locally rather than breaking the main code. <<

NOTE: this GitHub workflow is the BEST PRACTICE you should keep during a cooperative coding work. What we require from this project is that each of the group components should branch and merge at least once during the project.

